

WASP-52b

R-1874

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_210824 · created 2026-08-03T11:09:25+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459450.7352 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.06
Transit depth (Rp/Rs)^2	4.59 +/- 2.55 [%]
Orbital Inclination (inc)	83.23 +/- 2.66
Airmass coefficient 1 (a1)	1.03 +/- 0.072
Airmass coefficient 2 (a2)	-0.01 +/- 0.018
Scatter in the residuals of the lightcurve fit is	7.08 %
Best Comparison Star	#2 - [273, 470]
Optimal Aperture	3.48
Optimal Annulus	8.1
Transit Duration (day)	0.054 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.214 ± 0.06 vs 0.1646 (deviation 0.78σ)
Duration fit vs published	0.054 d vs 0.0758 d (deviation 0.69σ)
Mid-time O–C	15.8 min
Depth SNR	3.4
Residual scatter	7.08 %

Light curve

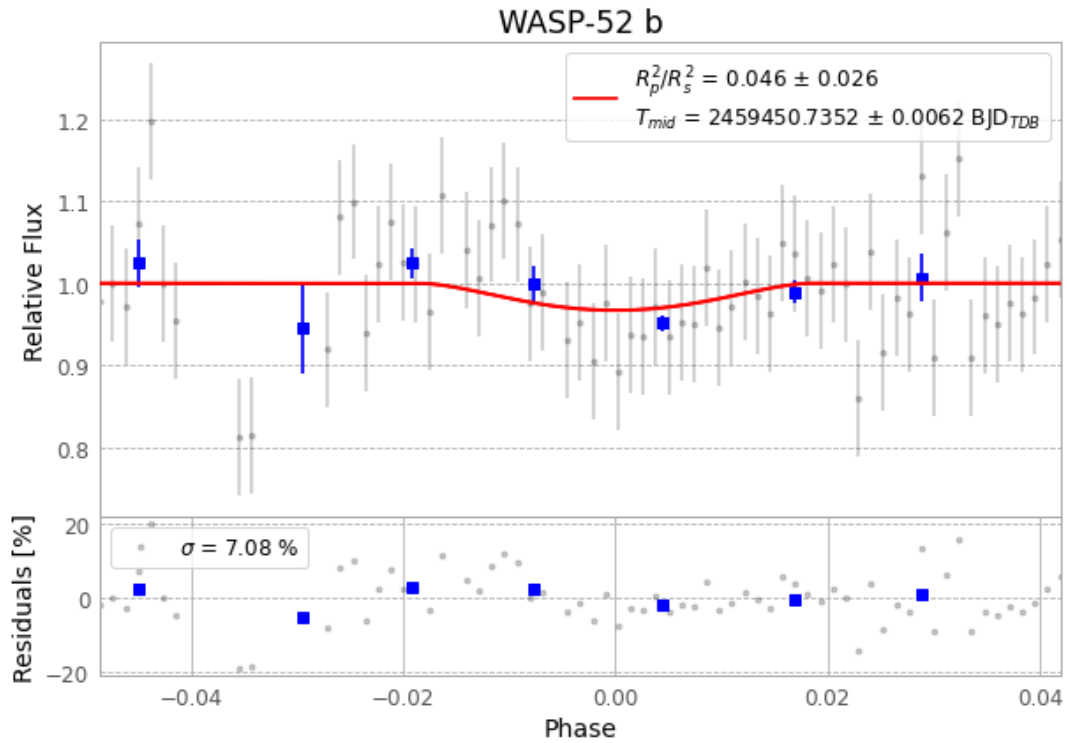


Figure 1 — FinalLightCurve_WASP-52 b_2021-08-23.png (SHA-256 524f6763164a5178...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [288, 157], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7b21dedd90828604...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52210824032712.FITS → WASP-52210824071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-210824.log (bf793192bdac...), FinalLightCurve_WASP-52 b_2021-08-23.png (524f6763164a...), FinalLightCurve_WASP-52 b_2021-08-23.csv (916de866359f...)

Compute resources

Wall time 300.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:09Z.